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compact1, compact2, compact3

java.nio.channels.spi

Class SelectorProvider

java.lang.Object
    java.nio.channels.spi.SelectorProvider

```
public abstract class SelectorProvider
extends Object
```

Service-provider class for selectors and selectable channels.

A selector provider is a concrete subclass of this class that has a zero-argument constructor and implements the abstract methods specified below. A given invocation of the Java virtual machine maintains a single system-wide default provider instance, which is returned by the `provider` method. The first invocation of that method will locate the default provider as specified below.

The system-wide default provider is used by the static `open` methods of the `DatagramChannel`, `Pipe`, `Selector`, `ServerSocketChannel`, and `SocketChannel` classes. It is also used by the `System.inheritedChannel()` method. A program may make use of a provider other than the default provider by instantiating that provider and then directly invoking the `open` methods defined in this class.

All of the methods in this class are safe for use by multiple concurrent threads.

Since:

1.4

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	`SelectorProvider()`	

	Initializes a new instance of this class.

Method Summary

Abstract Methods

Concrete Methods

Modifier and Type	Method	Descripti
Channel	inheritedChannel()	Returns the channel inherited from the entity th created this Java virtual machine
abstract DatagramChannel	openDatagramChannel()	Opens a datagran channel.
abstract DatagramChannel	openDatagramChannel(ProtocolFamily family)	Opens a datagran channel.
abstract Pipe	openPipe()	Opens a pipe.
abstract AbstractSelector	openSelector()	Opens a selector.
abstract ServerSocketChannel	openServerSocketChannel()	Opens a server-socket channel.
abstract SocketChannel	openSocketChannel()	Opens a socket channel.
static SelectorProvider	provider()	Returns the system-wide

		default selector provider for this invocatio of the Java virtual machine

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

SelectorProvider

```
protected SelectorProvider()
```

Initializes a new instance of this class.

Throws:

`SecurityException` - If a security manager has been installed and it denies `RuntimePermission("selectorProvider")`

Method Detail

provider

```
public static SelectorProvider provider()
```

Returns the system-wide default selector provider for this invocation of the Java virtual machine.

The first invocation of this method locates the default provider object as follows:

1. If the system property `java.nio.channels.spi.SelectorProvider` is defined then it is taken to be the fully-qualified name of a concrete provider class. The class is loaded and instantiated; if this process fails then an unspecified error is thrown.

2. If a provider class has been installed in a jar file that is visible to the system class loader, and that jar file contains a provider-configuration file named `java.nio.channels.spi.SelectorProvider` in the resource directory `META-INF/services`, then the first class name specified in that file is taken. The class is loaded and instantiated; if this process fails then an unspecified error is thrown.

3. Finally, if no provider has been specified by any of the above means then the system-default provider class is instantiated and the result is returned.

Subsequent invocations of this method return the provider that was returned by the first invocation.

Returns:

`The system-wide default selector provider`

openDatagramChannel

```
public abstract DatagramChannel openDatagramChannel()
                                                throws IOException
```

Opens a datagram channel.

Returns:

`The new channel`

Throws:

`IOException - If an I/O error occurs`

openDatagramChannel

```
public abstract DatagramChannel openDatagramChannel(ProtocolFamily family)
                                                throws IOException
```

Opens a datagram channel.

Parameters:

`family - The protocol family`

Returns:

`A new datagram channel`

Throws:

`UnsupportedOperationException - If the specified protocol family is not supported`

`IOException - If an I/O error occurs`

Since:

1.7

openPipe

```
public abstract Pipe openPipe()
                        throws IOException
```

Opens a pipe.

Returns:

The new pipe

Throws:

IOException - If an I/O error occurs

openSelector

```
public abstract AbstractSelector openSelector()
                                         throws IOException
```

Opens a selector.

Returns:

The new selector

Throws:

IOException - If an I/O error occurs

openServerSocketChannel

```
public abstract ServerSocketChannel openServerSocketChannel()
                                                      throws IOException
```

Opens a server-socket channel.

Returns:

The new channel

Throws:

IOException - If an I/O error occurs

openSocketChannel

```
public abstract SocketChannel openSocketChannel()
                                           throws IOException
```

Opens a socket channel.

Returns:

The new channel

Throws:

IOException - If an I/O error occurs

inheritedChannel

```
public Channel inheritedChannel()
                           throws IOException
```

Returns the channel inherited from the entity that created this Java virtual machine.

On many operating systems a process, such as a Java virtual machine, can be started in a manner that allows the process to inherit a channel from the entity that created the process. The manner in which this is done is system dependent, as are the possible entities to which the channel may be connected. For example, on UNIX systems, the Internet services daemon (*inetd*) is used to start programs to service requests when a request arrives on an associated network port. In this example, the process that is started, inherits a channel representing a network socket.

In cases where the inherited channel represents a network socket then the Channel type returned by this method is determined as follows:

- If the inherited channel represents a stream-oriented connected socket then a SocketChannel is returned. The socket channel is, at least initially, in blocking mode, bound to a socket address, and connected to a peer.
- If the inherited channel represents a stream-oriented listening socket then a ServerSocketChannel is returned. The server-socket channel is, at least initially, in blocking mode, and bound to a socket address.
- If the inherited channel is a datagram-oriented socket then a DatagramChannel is returned. The datagram channel is, at least initially, in blocking mode, and bound to a socket address.

In addition to the network-oriented channels described, this method may return other kinds of channels in the future.

The first invocation of this method creates the channel that is returned. Subsequent invocations of this method return the same channel.

Returns:

The inherited channel, if any, otherwise null.

Throws:

`IOException` - If an I/O error occurs

`SecurityException` - If a security manager has been installed and it denies `RuntimePermission("inheritedChannel")`

Since:

1.5

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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